Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 13

The Algebra of Sets

Covers Chartrand & Zhang, Section 2.2, Theorems 2.22 and 2.35

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

You are not learning new laws

The correspondence

Sets	Statements (Week 1)
$\cap$	$\wedge$
$\cup$	$\vee$
$\overline{}$	$\sim$
U	a tautology
$\emptyset$	a contradiction
$=$	$\equiv$

Every law in Theorem 2.22 and Theorem 2.35 is a Week 1 theorem read through this table.

Three mistakes to expect

1. **Negating the pieces but not the operation.** $\overline{A \cup B}$ is $\overline{A} \cap \overline{B}$ — the operation flips too.
2. **Treating a Venn diagram as a proof.** It shows one arrangement.
3. **Skipping the justification.** Each step of a derivation names a definition or a named theorem.

Part II: Practice Problems

Section A — Naming the source

1. For each set law, name the Week 1 result it comes from.

(a) $A \cup B = B \cup A$

(b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(c) $\overline{A \cap B} = \overline{A} \cup \overline{B}$

(d) $\overline{\overline{A}} = A$

Section B — Deriving

2. Derive $\overline{A} \cup B = \overline{A \cap \overline{B}}$, giving a reason for every step.

3. Use the laws — not a membership table — to simplify $\overline{\overline{A} \cup \overline{B}}$, naming each step.

Section C — Membership tables

4. Use a membership table to determine whether $A - (B \cup C) = (A - B) \cap (A - C)$.

A	B	C	$B \cup C$	$A - (B \cup C)$	$A - B$	$(A - B) \cap (A - C)$
T	T	T				
T	T	F				
T	F	T				
T	F	F				
F	T	T				
F	T	F				
F	F	T				
F	F	F				

Section D — Writing

5. In five to eight complete sentences, explain why the laws for sets did not need independent proofs once the laws for statements were established. What exactly is being translated, and what is doing the mathematical work?

Section E — Challenge (optional)

6. The correspondence table pairs U with a tautology and $\emptyset$ with a contradiction. Use that to predict, without computing, what each of these equals. Then check one of them.

(a) $A \cup U$ (b) $A \cap U$ (c) $A \cup \emptyset$ (d) $A \cap \emptyset$